

MuniAccountability™ — Integrated System Architecture & PRD

An Enterprise Product Specification by Carter Digitals (Pty) Ltd *Unified Platform Specification: AGSA Historical Audit Outcomes & National Treasury Live Financial Telemetry*

Document Metadata

Attribute	Details
Product	MuniAccountability™ (Standalone Platform)
Tagline	Every audit finding, alive between report cycles, powered by live financial telemetry.
Prepared by	Carter Digitals (Pty) Ltd — CIPC 2025/907839/07
Document Owner	Kabelo "Gee" Kadiaka, Founder & Director
Status	v3.0 — Core Integrated Architecture & Blueprint Spec
Date	July 2026
Classification	Commercial Confidential — Pitch Readiness Artifact

Table of Contents

1. Executive Summary & The Integration Paradigm
2. The Dual-Data Problem Statement
3. Core Product Vision & Architectural Pillars
4. User Personas & Data Contextualisation
5. Comprehensive Feature Specifications
6. Detailed Municipality API Architecture
7. System Information Architecture
8. Enriched Conceptual Data Model
9. End-to-End Technical Architecture & Data Ingestion Pipeline
10. User Experience, Interaction & Accessibility Principles
11. Compliance, Governance & Trust Matrix
12. Non-Functional Requirements & Service Level Agreements
13. Commercial Business Model & The ProtoLead Pitch Motion
14. Execution Roadmap & Phased Implementation

- 15. Appendix A: Comprehensive Glossary (AGSA & Treasury Cubes)
- 16. Appendix B: Open Questions for Strategic Validation

1. Executive Summary & The Integration Paradigm

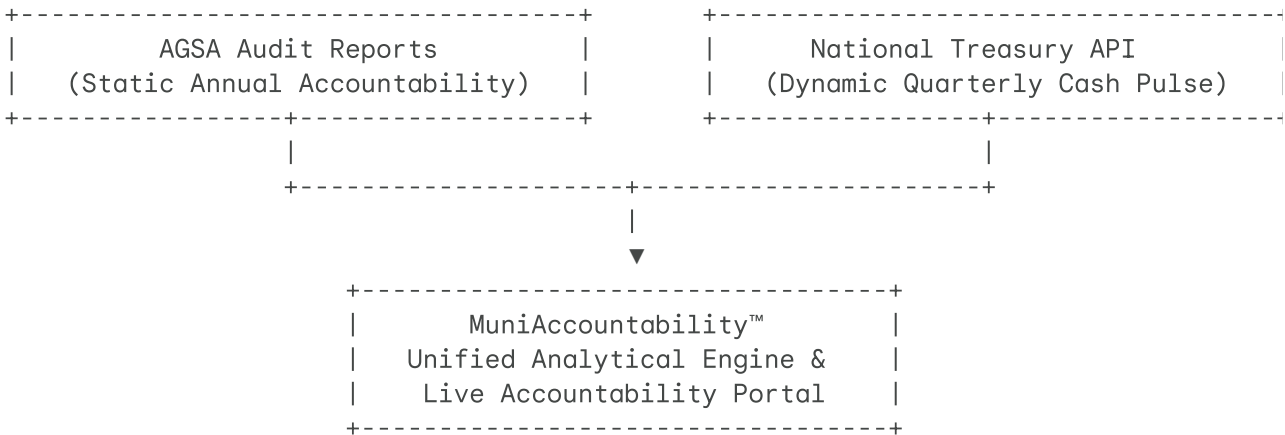
1.1 Context

Every fiscal year, the Auditor-General of South Africa (AGSA) releases a consolidated audit report covering all 257 municipalities across the nation. This report acts as the definitive forensic record of public funds management, institutional governance, and compliance. Concurrently, the National Treasury maintains the Municipal Money portal, which aggregates statutory municipal financial submissions—including budgets, quarterly expenditure adjustments, and cash flow reports—into an operational data ecosystem accessible via a JSON API.

Historically, these two datasets have remained structurally isolated. AGSA’s authoritative conclusions are published as static, 300-page PDF documents that lack temporal flexibility. Meanwhile, National Treasury’s raw API tables provide continuous numerical streams but lack the governance context, compliance citations, and legal gravity necessary to drive institutional accountability.

1.2 The Innovation

MuniAccountability bridges this gap by merging these two data vectors into a single, cohesive, standalone platform. The platform ingests historical audit findings and synthesises them directly with live, quarterly telemetry from the Municipal Money API cubes (`incexp` , `cflow` , `bsheet` , `capital` , `conditional_grants`).



By overlaying dynamic operational spending directly onto historical governance baselines, MuniAccountability transforms a backward-looking annual archive into an early-warning diagnostic application. The platform operates independently of internal government suites, serving as a dedicated transparency engine for public oversight bodies, provincial authorities, civil researchers, media houses, and active civic institutions.

2. The Dual-Data Problem Statement

2.1 The Friction of Static Accountability Data

According to the 2024-25 AGSA Consolidated General Report on Local Government Audit Outcomes, local government governance faces significant systemic challenges:

- **Minimal Excellence:** Only 39 out of 257 municipalities (15%) achieved clean audits, representing a mere 8% of total local government expenditure.
- **Active Regression:** 38 municipalities (15%), managing 24% of the collective budget allocation, have regressed in performance since the 2020-21 cycle—a group that includes three major metropolitan councils.
- **Chronic Failures:** 8 municipalities received disclaimed opinions (the worst possible outcome, indicating an inability to present auditable books), with 7 of these councils trapped in repeat disclaimers for 3 to 10 consecutive financial cycles.
- **The Compliance Illusion:** The 117 councils classified as "unqualified with findings"—technically a passing grade—were responsible for 45% of all irregular expenditure, 37% of unauthorised expenditure, and 31% of reported material irregularities nationwide.
- **Enforcement Separation:** AGSA issued its first-ever personal certificate of debt to compel a municipal manager to personally repay public financial losses. However, no public system currently tracks the legal lifecycle, enforcement status, or legal precedent of these directives.

When this data is locked inside static text layers, it remains hidden from active public oversight and policy intervention.

2.2 The Friction of Raw Financial Feeds

National Treasury provides raw public access to municipal financial data through the `municipaldata.treasury.gov.za/api` endpoints. However, this dataset presents distinct usability hurdles:

- **Absence of Context:** A raw database query showing an operating deficit in the `incexp` cube does not state whether the deficit stems from a structural crisis, an unresolved Material Irregularity, or a failure to implement consequence management.
- **High Technical Friction:** Accessing data across multiple cubes requires advanced knowledge of data aggregation, JSON parsing, and fiscal structural keys. This creates a barrier for non-technical analysts, civil society actors, and journalists.
- **Delayed Insight:** Raw numerical rows do not inherently highlight correlations, trends, or institutional risk trajectories until external data teams extract and evaluate them.

2.3 The Solution: Contextualized Synthesis

MuniAccountability addresses these structural challenges by mapping live numerical flows against established compliance benchmarks:

Target User Group	Current Status Quo	MuniAccountability Intervention
-------------------	--------------------	---------------------------------

Provincial Oversight Analysts	Manually cross-referencing past PDFs with current Treasury spreadsheets prior to field assessments.	A single, unified dashboard combining the historical AGSA audit trail with current-quarter cash runways.
Municipal CFOs & Officers	Lack direct, automated comparison tools to benchmark institutional progress against peer structures.	Dynamic benchmarking tools that compare internal budget variations directly against clean-audit peer councils.
Investigative Journalists	Manually compiling annual data stories from disparate public text records.	Programmatic API search parameters that instantly flag repeat disclaimer trends and long-term unresolved MLs.
Active Ratepayers & Residents	Left with no clear, low-bandwidth access point to review neighborhood administrative compliance.	A simplified public interface providing plain-language summaries and trend indicators, completely free of administrative jargon.

3. Core Product Vision & Architectural Pillars

3.1 Vision Statement

MuniAccountability delivers a direct view into municipal financial compliance and operational trajectories. It is designed to be accessible to a wide audience, providing key institutional insights within 10 seconds, while offering robust downstream data access through a comprehensive API.

3.2 Main Structural Pillars

MuniAccountability™ Core Operational Pillars		
1. RADICAL ACCESSIBILITY	Visual, plain-language UI	
2. REAL-TIME TELEMETRY	Treasury API integration	
3. STRICT TRACEABILITY	Verification via citations	
4. PROGRAMMATIC UTILITY	Robust downstream API	

- Radical Accessibility:** Financial datasets are paired with plain-language, contextual explanations. Complexity is organized hierarchically through drill-down paths, keeping the primary user interface clean and legible.
- Real-Time Telemetry:** The platform complements annual retrospective audits by integrating live data from Treasury API cubes, showing current spending behaviors between formal report releases.
- Strict Traceability:** To maintain historical reliability, every audit-derived figure contains a clear citation pointing directly back to the original source document page.

4. **Programmatic Utility:** The downstream API converts raw public data into reliable infrastructure, allowing media houses, academic environments, and external applications to build on top of a unified data source.

4. User Personas & Data Contextualisation

4.1 Persona 1: "Nomvula" — Provincial Oversight Analyst

- **Objective:** Compiling briefing dossiers for structural reviews and field oversight visits.
- **Friction:** Spends significant time manually consolidating past audit comments with recent municipal cash flow reports.
- **Platform Workflow:** Downloads a unified, single-page profile containing the five-year AGSA compliance history paired with current-quarter conditional grant utilization rates.

4.2 Persona 2: "Thabo" — Municipal CFO

- **Objective:** Monitoring internal performance compliance and tracking corrective actions before formal audits.
- **Friction:** Struggles to effectively benchmark current expenditure metrics against equivalent municipal structures.
- **Platform Workflow:** Accesses a secure peer dashboard to track the council's current budget variances against a cohort of clean-audit municipalities within the same budget tier.

4.3 Persona 3: "Aisha" — Investigative Journalist

- **Objective:** Identifying structural governance failures, systemic waste, and unresolved financial irregularities.
- **Friction:** Navigates cumbersome public document archives to connect historical failures with current operational trends.
- **Platform Workflow:** Sets up structured queries via the public developer tier to isolate councils with multiple outstanding Material Irregularities that also show deteriorating cash reserves.

4.4 Persona 4: "Sipho" — Policy Analyst

- **Objective:** Identifying early performance regressions to inform targeted provincial interventions.
- **Friction:** Limited to reviewing historical data sheets, which often delays preventative policy updates.
- **Platform Workflow:** Reviews the National Trends dashboard to monitor councils whose current operational expenditure deviations suggest potential long-term audit status regressions.

4.5 Persona 5: "Ratepayer / Community Resident"

- **Objective:** Checking local municipality performance clearly without navigating complex accounting terms.
- **Friction:** Confronted by dense administrative terminology and confusing public balance sheets.

- **Platform Workflow:** Enters a postal code into an open search bar to instantly view a plain-language summary card detailing local council audit trends, active challenges, and basic spending performance.

4.6 Persona 6: "Civic Developer / Systems Partner"

- **Objective:** Incorporating reliable local governance metrics directly into research dashboards and civic tools.
- **Friction:** Processing unformatted public text records and managing erratic raw data sources manually.
- **Platform Workflow:** Registers for public API access keys to stream clean, pre-structured municipal performance data directly into external platforms.

5. Comprehensive Feature Specifications

5.1 The Unified Municipality Profile Screen

CITY OF TSHWANE METROPOLITAN MUNICIPALITY					
[Status: Unqualified with Findings]		[Trajectory: Stable —]			

LIVE PULSE: Cash Runway: 1.4 Months		Capital Budget Ingest: 42%			

INTERACTIVE HISTORY TIMELINE					
2021 [Clean]		-> 2022 [Clean]		-> 2023 [Unqual] -> 2024 [Unqual with Find]	

ACTIVE MATERIAL IRREGULARITIES (4)					
- MI-024: Fleet Procurement Fraud		R14.2M	Status:	Investigation	
- MI-027: Substation Underspend		R45.0M	Status:	Remediation	

LIVE BUDGET MONITOR (Treasury API Feed)					
Operating Budget Burn:			68% (Q3 Target: 75%)		
Grant Drawdown Risk:			32% (CRITICAL UNDERSPEND)		

The profile screen provides a comprehensive view of a municipality's financial and operational health by combining historical audit data with live financial metrics. Each municipality is assigned a permanent URL (</muniaccountability.co.za/municipalities/{slug}>).

- **Headline Status Card:** Displays the current AGSA audit outcome category, a calculated trajectory direction, and an automated plain-language summary of the council's current operational standing.
- **Interactive History Timeline:** A multi-year visual timeline mapping historical audit opinions. Users can click on individual years to view specific audit parameters and performance flags.
- **Material Irregularities Lifecycle Framework:** A live operational matrix detailing active Material Irregularities, tracking their current stage (Investigation, Referral, Remediation, Debt Certificate Process), and displaying total financial exposure.

- **Live Cash Runway Component:** Computes current operational liquidity by processing cash and cash-equivalent values from the Treasury `cflow` and `bsheet` data cubes:

$$\text{Cash Runway (Months)} = \frac{\text{Cash \& Cash Equivalents (from cflow / bsheet)}}{\text{Average Monthly Operational Outflow (from incexp)}}$$

- **Capital & Conditional Grant Tracking Gauge:** Integrates live data from the `conditional_grants` cube to display allocations alongside real-time quarterly drawdowns, automatically flagging under-spending on critical infrastructural grants.
- **Comparative Cohort Tool:** Automatically evaluates a municipality's financial efficiency against equivalent public structures within the same budget band and geographical category.
- **Source Citation System:** Provides transparent page references for all historical entries, linking data points back to the official source documents.

5.2 National & Provincial Analytics Map

- **Geographic Visualisations:** A responsive, interactive national map color-coded by current audit status classifications. Users can zoom from national overviews down to specific provincial regions and local municipal boundaries.
- **Provincial Aggregations:** Displays consolidated operational indicators by province, including total expenditure risks, active disclaimers, and overall regional performance trends.
- **Performance Tracking Lists:** Identifies and ranks municipalities showing notable improvements or significant budget variances relative to preceding cycles.

5.3 High-Risk Radar & Disclaimer Tracking

- **Chronic Overrun Flags:** Automatically isolates and highlights any council that has received a Disclaimed or Adverse opinion for two or more consecutive fiscal years.
- **Oversight Dossier Generator:** Allows users to export structured operational briefings that summarize multi-year compliance failures, helping teams reference relevant legislative intervention frameworks.
- **Provincial Resolution Trackers:** Monitors the reduction or persistence of critical audit categories within individual provinces across a user-defined multi-year baseline.

5.4 Material Irregularity Lifecycle Engine

- **Progress Tracking Systems:** Tracks individual Material Irregularities through their entire legal lifecycle, from identification to formal recovery actions.
- **Age Resolution Alerts:** Automatically highlights unresolved items that have remained open beyond standard institutional resolution windows (default set to 24 months).
- **Systemic Recovery Analytics:** Computes overall recovery performance by comparing total financial losses against actual recovered amounts across national, provincial, and local levels.

5.5 Financial Health & Budget Variance Analytics

- **Historical Leakage Visualisations:** Tracks irregular, unauthorized, and wasteful expenditure metrics over time, showing performance trends across consecutive fiscal cycles.
- **Variance Evaluation Tools:** Compares historical audit findings with live budget adjustments from the Treasury `incexp` cube to identify patterns of operational stress.

5.6 Infrastructure Project Risk Module

- **Grant Utilization Analysis:** Tracks infrastructure project performance by evaluating grant allocations against actual project completion milestones.
- **Maintenance Risk Identification:** Evaluates capital maintenance spending against total asset values using the `capital` cube to identify potential maintenance backlogs:

$$\text{Maintenance Spend Ratio} = \frac{\text{Asset Maintenance Expenditure (capital)}}{\text{Total Property, Plant \& Equipment (bsheet)}}$$

- **Cross-Reference Functions:** Links operational infrastructure project delays with corresponding audit findings and Material Irregularities.

5.7 Metropolitan Administration Spotlight

- **Dedicated Metro Cohort Dashboard:** A specialized analytical interface for South Africa's 8 metropolitan councils.
- **Core Performance Indicators:** Compares metropolitan performance across service delivery metrics, revenue collection, infrastructure investment, and compliance frameworks.

5.8 Administrative Accountability Scorecard

- **Resolution Tracking:** Tracks institutional compliance by comparing recommended disciplinary actions against actual implementation.
- **Role-Level Records:** Logs enforcement outcomes and legal precedents at the organizational role level to preserve institutional memory.

5.9 Public Search Engine

- **Flexible Search Parameters:** A fast search function allowing users to locate profiles by municipality name, province, or postal code.
- **Unauthenticated Public Tier:** Provides open access to standard public profiles and plain-language summaries without requiring user registration.

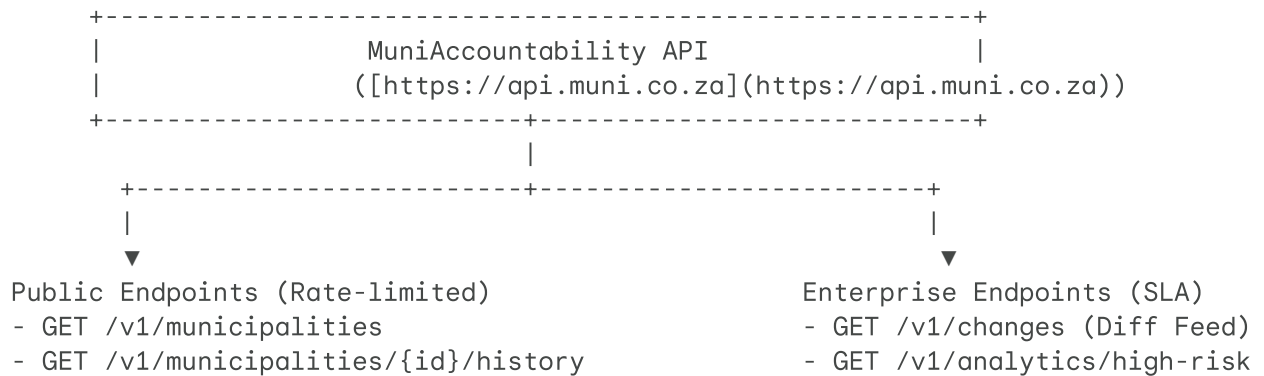
5.10 Automated Performance Briefing Exports

- **One-Click Document Engine:** Generates formatted PDF or Word briefing documents for individual municipalities or provincial overviews.
- **Structured Templates:** Produces standard report layouts designed for council sessions, public oversight updates, or journalistic reporting.

6. Detailed Municipality API Architecture

6.1 Downstream API Topology

The platform provides access to its structured data ecosystem via the dedicated downstream API service (/v1/). This allows partner organizations, researchers, and media outlets to systematically query integrated municipal data.



6.2 Primary Operational Endpoints

GET /v1/municipalities

- **Description:** Retrieves a filterable list of all 257 municipalities.
- **Parameters:**
 - `province` (string, optional): e.g., "Gauteng"
 - `category` (string, optional): e.g., "metro" , "local" , "district"
 - `current_outcome` (string, optional): e.g., "clean" , "disclaimer"
 - `page` (integer, default: 1)
- **Response Prototype (200 OK):**

```
{
  "status": "success",
  "meta": {
    "total_records": 257,
    "page": 1,
    "per_page": 20
  },
  "data": [
    {
      "id": "ZA_MET_TSH",
      "name": "City of Tshwane Metropolitan Municipality",
      "province": "Gauteng",
      "category": "metro",
      "budget_tier": "Tier_1",
      "integrity_metrics": {
        "current_audit_outcome": "Unqualified with Findings",
        "trajectory": "stable",
        "active_material_irregularities": 4
      }
    }
  ]
}
```

```

    },
    "live_financial_pulse": {
      "cash_runway_months": 1.4,
      "operating_budget_burn_percentage": 68.2,
      "grant_drawdown_risk_level": "CRITICAL"
    },
    "links": {
      "profile": "[https://muniaccountability.co.za/municipalities/city-of-tshwar
    }
  }
]
}

```

GET /v1/municipalities/{id}/integrity-health

- **Description:** Returns a detailed compliance record for a specific municipality, combining historical audit outcomes with live cash-runway metrics.
- **Response Prototype (200 OK):**

```

{
  "municipality_id": "ZA_MET_TSH",
  "fiscal_year_reference": "2025-2026",
  "historical_baseline": {
    "audit_outcome_trail": [
      { "year": "2024-25", "outcome": "Unqualified with Findings", "source_page": 1
      { "year": "2023-24", "outcome": "Unqualified with Findings", "source_page": 1
      { "year": "2022-23", "outcome": "Clean Audit", "source_page": 98 }
    ]
  },
  "live_telemetry_cube_aggregates": {
    "cflow_reported_cash_balance": 145000000.00,
    "incexp_monthly_operating_burn": 103571428.57,
    "calculated_cash_runway_months": 1.4,
    "conditional_grants_drawdown": {
      "allocated_infrastructure_grants": 450000000.00,
      "spent_to_date_q3": 144000000.00,
      "drawdown_variance_percentage": -43.0
    }
  },
  "active_material_irregularities": [
    {
      "mi_id": "MI-2025-024",
      "description": "Systemic irregularities in fleet procurement contracts",
      "financial_exposure": 14200000.00,
      "current_stage": "Investigation",
      "days_unresolved": 310,
      "source_citation": "AGSA MFMA 2024-25, Page 144"
    }
  ]
}

```

GET /v1/analytics/high-risk-grants

- **Description:** Identifies infrastructure projects showing high financial risk, where historical audit findings overlap with lower-than-expected spending rates in the current quarter's conditional_grants data.
- **Response Prototype (200 OK):**

```
{
  "generated_at": "2026-07-04T23:45:00Z",
  "risk_cohort_count": 14,
  "high_risk_infrastructure_records": [
    {
      "municipality_id": "ZA_MP_MP326",
      "municipality_name": "City of Mbombela Local Municipality",
      "historical_infrastructure_finding_flag": true,
      "grant_type_reference": "MIG (Municipal Infrastructure Grant)",
      "live_metrics": {
        "annual_allocation": 189000000.00,
        "q3_actual_expenditure": 42000000.00,
        "expected_expenditure_benchmark": 141750000.00,
        "spending_deficit": 99750000.00,
        "calculated_drawdown_efficiency_ratio": 22.2
      },
      "risk_classification": "CRITICAL_UNDERSPEND"
    }
  ]
}
```

GET /v1/changes

- **Description:** A data feed listing system records modified since a user-specified timestamp. Designed for synchronization by enterprise applications and partner tools.
- **Parameters:** since (ISO 8601 string, required)

6.3 Programmatic Access Structure

Tier Name	Target Group	Allocation Parameters	Data Scope
Public Developer	Academic researchers, independent hobbyists, civic tech testers.	Rate limit set to 60 queries/minute. Free tier access.	Open public data fields, historical records, and standard profile metrics.
Partner Media	News organizations, publishing networks, civil society NGOs.	Tiered subscription structure. Expanded rate limits with webhook functionality.	Comprehensive access, bulk export options, change notifications, and full historical datasets.

Institutional	Provincial government divisions, national review boards, rating entities.	Enterprise service level agreement with high uptime guarantees and custom endpoints.	Unrestricted access across all operational data fields, including system audit histories.
----------------------	---	--	---

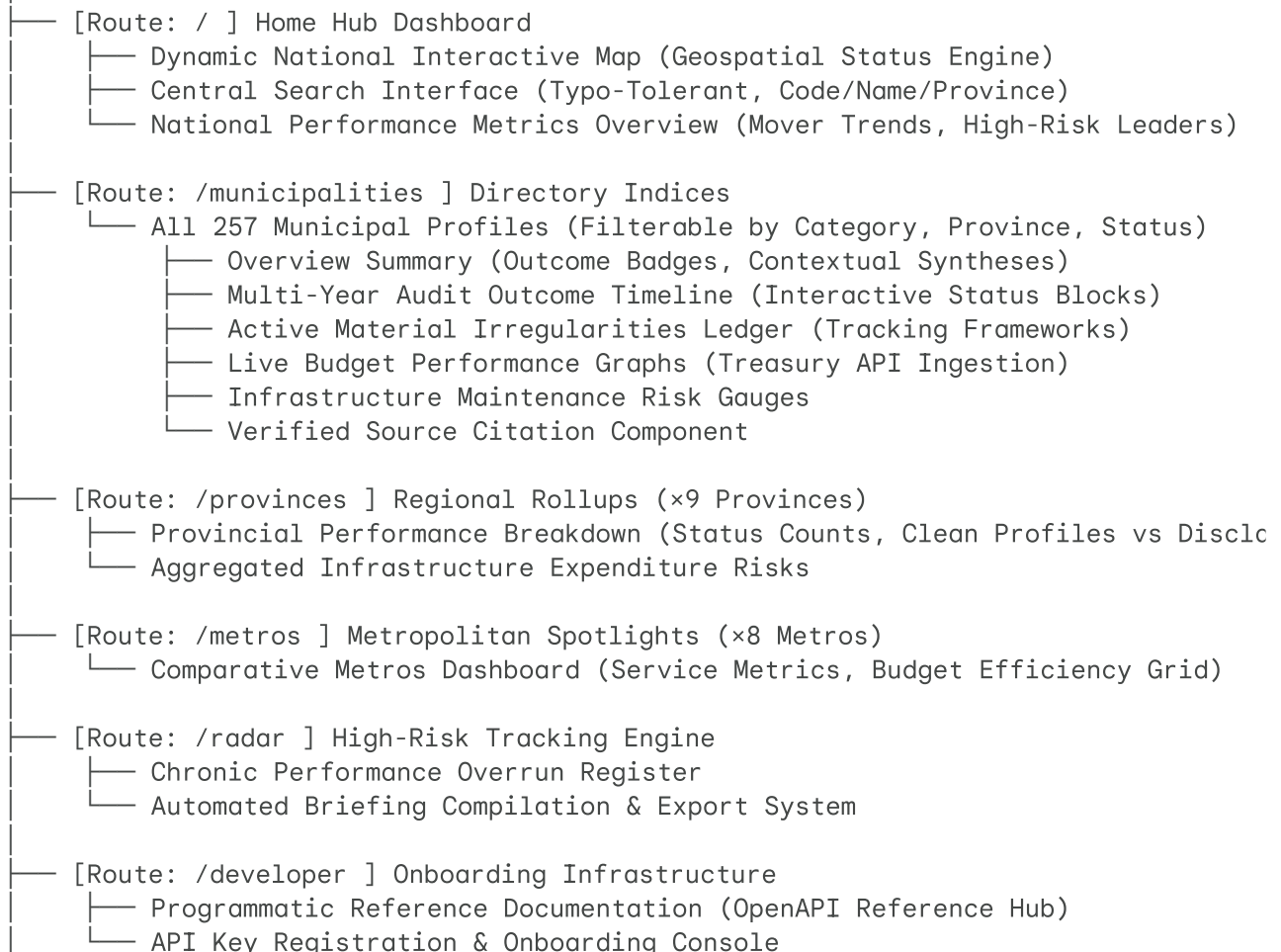
6.4 API Design Standards

- **Citations by Design:** All profile and analytical endpoint payloads include explicit source citation properties, ensuring downstream applications maintain data verification.
- **Stable Versioning Strategy:** Endpoints run on the `/v1/` route path. Backward compatibility is maintained through structured deprecation cycles before any architectural shifts.
- **Automated Onboarding Portal:** Public developer keys are provisioned through an automated online onboarding dashboard to support civic and academic tool developers.
- **Standard Reference Documentation:** The downstream API features comprehensive, interactive OpenAPI/Swagger reference manuals with functional code examples.

7. System Information Architecture

The platform architecture is structured to allow users to navigate directly from high-level national indicators down to specific regional details and source documentation.

MuniAccountability™ Standalone Core Web Shell



- `id` (String, Primary Key): Standard format identifier, e.g., "ZA_MET_TSH" .
- `name` (String): Official administrative title, e.g., "City of Tshwane Metropolitan Municipality" .
- `province` (String): Regional assignment, e.g., "Gauteng" .
- `category` (Enum): Options include `metro` , `local` , or `district` .
- `budget_band` (Enum): Tiers graded from `Tier_1` (large metros) down to `Tier_4` (small rural councils).
- `created_at` (Timestamp): System record generation date.

AuditCycle (Historical AGSA Dimension)

- `id` (String, Primary Key): Unique row hash.
- `municipality_id` (String, Foreign Key -> `Municipality.id`).
- `fiscal_year` (String): e.g., "2024-2025" .
- `outcome` (Enum): Options include `Clean` , `Unqualified_Findings` , `Qualified` , `Adverse` , or `Disclaimer` .
- `prior_year_outcome` (Enum): Outcome of the preceding cycle.
- `movement_flag` (Enum): Tracked as `Improving` , `Regressing` , or `Stable` .
- `expenditure_represented` (Decimal): Total budget value validated during the audit.
- `source_page_ref` (Integer): Exact page number reference within the official report document.

MaterialIrregularity (Historical Tracker Entity)

- `id` (String, Primary Key): Format code, e.g., "MI-2025-024" .
- `municipality_id` (String, Foreign Key -> `Municipality.id`).
- `fiscal_year_raised` (String): Operational year of identification.
- `category` (String): Type classification, e.g., "Procurement Non-Compliance" .
- `financial_value` (Decimal): Total calculated financial exposure.
- `current_status` (Enum): Options include `Raised` , `Under_Investigation` , `Referred_Enforcement` , `Remediation` , or `Resolved` .
- `status_history` (Array[JSON]): Log entries capturing dates and state transitions over time.
- `recovery_amount` (Decimal): Funds clawed back during remediation actions.
- `source_page_ref` (Integer): Exact page number reference within the official report document.

ExpenditureRecord (Combined Analysis Entity)

- `id` (String, Primary Key): Combined unique identifier hash.
- `municipality_id` (String, Foreign Key -> `Municipality.id`).
- `fiscal_year` (String): e.g., "2025-2026" .
- `irregular_amount` (Decimal): Historical audit baseline metric.
- `unauthorised_amount` (Decimal): Historical audit baseline metric.
- `fruitless_wasteful_amount` (Decimal): Historical audit baseline metric.
- `live_quarterly_variance` (Array[JSON]): Current quarterly monitoring metrics parsed from the Treasury API `incexp` cube.

InfrastructureProject (Treasury API Alignment Entity)

- `id` (String, Primary Key): Unique project identifier.
- `municipality_id` (String, Foreign Key -> `Municipality.id`).
- `grant_type` (String): Treasury data cube alignment field, e.g., "MIG" .

- `allocated_amount` (Decimal): Standard funding allotment value from the `conditional_grants` data stream.
- `spent_amount` (Decimal): Real-time expenditure values retrieved from the current financial period check.
- `completion_status_percentage` (Float): Physical progress status indicators where available.
- `maintenance_risk_flag` (Boolean): Calculated alert condition determined by parsing structural indices from the `capital` and `bsheet` data cubes.

EnforcementAction (Precedent Monitoring Entity)

- `id` (String, Primary Key): System tracker code.
- `municipality_id` (String, Foreign Key -> `Municipality.id`).
- `action_type` (Enum): Options include `Certificate_Of_Debt`, `Disciplinary_Recommendation`, or `Criminal_Referral`.
- `target_official_role` (String): Role-level tracking identifier, e.g., "Municipal Manager".
- `current_stage` (Enum): Options include `Issued`, `Contested`, `Payment_In_Progress`, or `Enforced`.
- `source_page_ref` (Integer): Exact page number reference within the official report document.

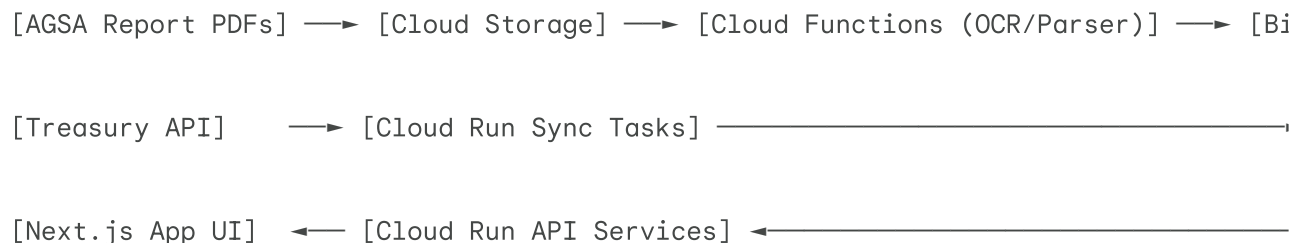
SourceDocument (System Processing Entity)

- `id` (String, Primary Key): Source record identifier.
- `report_year` (String): Operational reporting period.
- `publisher` (String): Originating authority, typically "AGSA".
- `storage_uri` (String): Secure internal cloud file reference path.
- `ingestion_status` (Enum): Options include `Staged`, `Processing`, `Verified`, or `Deployed`.

9. Technical Architecture & Data Ingestion Pipeline

9.1 Platform Stack Design

The platform uses an independent cloud architecture designed to isolate web application tasks from processing-heavy database activities.

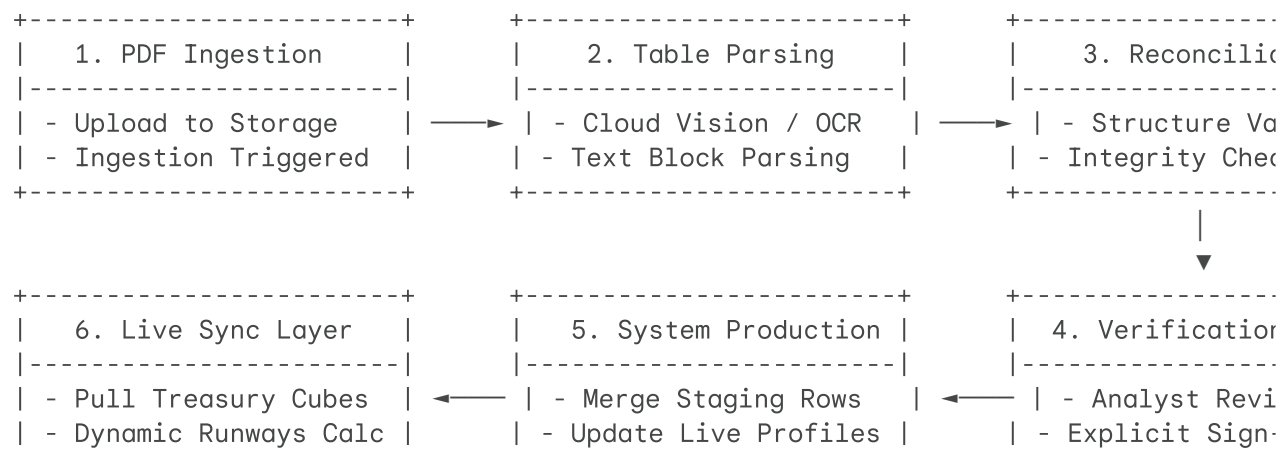


- **User Interface Layer:** Next.js using the App Router framework. Renders pages server-side to maintain responsive delivery times and support search visibility indexing, while handling interactive chart elements client-side.
- **Identity & Access Controls:** Firebase Authentication manages platform entry, separating basic public user paths from enterprise partner access configurations.
- **Operational Key-Value Storage:** Cloud Firestore manages configuration metrics, user portfolio preferences, API authorization allocations, and basic administrative updates.
- **Analytical Storage Engine:** BigQuery serves as the primary system repository, managing historical records and incoming data streams while processing cross-cube queries asynchronously.
- **Independently Scalable Services Layer:** Cloud Run acts as the primary service host, isolating public interface activities from systemic downstream data query volumes.
- **Document Ingestion Engine:** Cloud Functions orchestrate PDF document parsing tasks, extracting structured metrics into analytical staging environments for verification before deployment.
- **Secure File Hosting:** Cloud Storage acts as the platform repository for original public documents and dynamically generated briefing packages.
- **Generative Text Layer:** Anthropic Claude API integration provides plain-language descriptive insights based on raw numerical variances, operating under pre-configured system wording rules.
- **Infrastructure Automation Tools:** Deployment tasks are configured through independent Terraform codebases, ensuring isolated platform resource management.

9.2 Ingestion & Processing Standards

- **Asynchronous Database Operations:** Database executions, reporting compilations, and API query distributions are run as non-blocking asynchronous tasks. This ensures page responsiveness remains independent of background query volumes.
- **Regional Deployment Footprint:** System components are hosted within the Google Cloud Platform `africa-south1` (Johannesburg) data infrastructure to align with local data management conventions.

9.3 Ingestion Pipeline Workflow



- +-----+ +-----+ +-----
- Document Loading:** Raw audit report assets are deposited into secure storage environments, triggering processing workers via automated notification systems.
 - Text & Table Isolation:** Document processing routines isolate tabular blocks and textual statements, converting narrative data into clean, queryable key-value records.
 - Data Mapping:** Extracted fields are mapped against established municipal classification models, validating that records match baseline formatting standards.
 - Verification Environment:** Incoming records are sent to a restricted staging dashboard where an administrator verifies formatting consistency and confirms the accuracy of extracted data before deployment.
 - Production Deployment:** Verified data rows are moved to active analytics tables, updating live municipality profiles across the public web front-end.
 - Live Data Integration:** The platform synchronizes with Treasury API data cubes (`incexp` , `cflow` , `bsheet` , `capital` , `conditional_grants`) on a quarterly schedule. The engine recalculates metrics like cash runways and grant utilization rates, providing an updated operational summary ahead of the next formal audit release.

10. User Experience, Interaction & Accessibility Principles

10.1 Presentation Guidelines

- Contextual Data Display:** Visual indicators and numerical values are accompanied by brief plain-language summaries, helping users understand performance metrics without needing a background in municipal finance.
- Fluid Navigation Paths:** Dynamic interface elements support intuitive interactions, allowing users to select visual items to view granular supporting details and citations.
- Direct Verification Paths:** Critical information entries provide direct access to original source details, allowing users to verify facts within two clicks.
- Mobile-First Performance Standards:** The public tier interface is optimized for mobile performance, aiming for sub-2-second load times on standard 3G mobile connections to support broad citizen access.
- Inline Explanations:** System pages feature contextual definitions for specialized administrative terms (e.g., *Disclaimed Opinion*, *Material Irregularity*) to maintain clarity across all user groups.

10.2 Component Presentation Matrix

Target Document Hierarchy & Typography Style Guide		
Heading 1: Project Identity	22pt Bold, Primary Deep Slate	
Heading 2: Component Title	14pt Bold, Left Border Highlight	

Heading 3: Item Classification	11pt Italic, Muted Grey Accent	
Body Text: Standard Metrics	10pt Regular, Sans-Serif Clear	

- **Heading 1 (Main Project Identifiers):** Set to 22pt Bold using a deep slate tone to provide a clear visual anchor.
- **Heading 2 (System Component Titles):** Set to 14pt Bold, featuring an off-side color accent border to clearly mark section breaks.
- **Heading 3 (Specific Classification Items):** Set to 11pt Italic in a soft charcoal tone to establish clear layout hierarchy.
- **Body Text (Standard Interface Entries):** Set to 10pt Regular in an open sans-serif layout to maintain text readability across small mobile screens.

11. Compliance, Governance & Trust Matrix

11.1 Regulatory and Operational Alignment

- **Protection of Personal Information Act (POPIA) Alignment:** System tracking parameters focus strictly on official institutional roles (e.g., *Municipal Manager, Chief Financial Officer*) rather than individual personal records. Enforcement data entries use public records published directly by AGSA, ensuring compliance with local personal information protections.
- **Systemic Citation Requirements:** The analytical data schema enforces the inclusion of source file tracking parameters for all historical records, ensuring data reliability.
- **Content Generation Wording Rules:** Automated summary routines use neutral descriptive language (e.g., *Unresolved Status for 24 Months, Budget Variance Deviation*) rather than legally complex labels, except when directly quoting published source text. This approach ensures narrative clarity while protecting platform credibility.
- **Data Rights Management:** The platform aggregates public facts, figures, and tracking references from open public reports, rather than reproducing copyrighted text blocks. Data distribution configurations are periodically reviewed to ensure compliance with open access provisions.
- **Methodology Transparency Core:** A dedicated explanation page details validation steps, processing rules, and ingestion logs, ensuring the platform remains accountable to its users.

12. Non-Functional Requirements & Service Level Agreements

12.1 Performance Benchmarks

- **Data Delivery Metrics:** Basic public profile screens are optimized to render on standard mobile connections within 2 seconds.
- **System Accessibility Standard:** Production environments are configured to deliver 99.5% availability for public search features, and 99.9% availability for enterprise programmatic API channels under service level agreements.

- **Interface Protection Measures:** Access pathways include active rate-limiting and web application firewalls to protect system integrity from automated scraping attempts.
- **Operational Validation Logging:** Ingestion updates, system record state modifications, and administrative verifications are captured within unalterable audit tracking frameworks.
- **Universal Interface Accessibility:** Front-end designs follow WCAG 2.1 AA design standards to support accessible text scaling and screen reader compatibility.
- **Localized Resource Footprint:** Production data operations run within localized regional computing zones to comply with regional data governance practices.
- **Resource Scaling Controls:** The programmatic distribution layer runs on micro-services that scale down when inactive, minimizing ongoing infrastructure costs.
- **Cost Efficiency Parameters:** The MVP architecture runs within low-cost operational tiers, utilizing auto-scaling micro-services and on-demand database querying to control costs before commercial deployment.

13. Commercial Business Model & The ProtoLead Pitch Motion

13.1 Monetisation Streams

1. **Institutional Access Models:** Annual software licensing packages designed for provincial oversight bodies, oversight committees, and national tracking divisions requiring deep analytical access.
2. **Programmatic Data Subscriptions:** Tiered api usage structures for research teams, non-governmental groups, and media organizations requiring bulk data access or high-volume query limits.
3. **Open Access Public Tier:** A free, public web interface that provides broad civil access while serving as an operational showcase for institutional and enterprise data tiers.
4. **Custom Briefing Delivery:** Targeted reporting options allowing organizations to request tailored briefing portfolios for local governance evaluations or public oversight cycles.

13.2 The ProtoLead Pitch Strategy

The platform uses functional, high-fidelity prototypes of target districts to engage potential institutional partners.

The ProtoLead Pitch Cycle

1. CONSTRUCT CURATED PROTOTYPE	
Ingest 3-year baseline for a single target province.	
2. DYNAMIC LIVE DEMONSTRATION	
Present live grant drawdown risk during meetings.	

Rather than pitching via generic presentation decks, engagement teams demonstrate live municipal profiles showing a region's current conditional grant drawdown performance. Showing real-time resource risks provides immediate, practical value to public oversight teams, changing the engagement from a standard software presentation into an active operational assessment.

14. Execution Roadmap & Phased Implementation

Phase 0: Foundation	Phase 1: MVP Release	Phase 2: Depth Expansion
[-----]	[-----]	[-----]
- 5-Year Data Prep	- Public Map Portal	- Full Treasury Sync
- Schema Baseline	- Core API v1 Tier	- Briefing Generator
- Parser Automations	- Radar Alerts Setup	- Custom Webhooks Feed

14.1 Phase 0: Foundations & Data Preparation

- **Deliverables:** Ingest a five-year baseline of historical report data, finalize primary data schemas, deploy isolated cloud sandboxes, and verify primary document parsing scripts.
- **Outcome:** A verified, production-ready historical analytical dataset.

14.2 Phase 1: Core Functional MVP Release

- **Deliverables:** Launch the primary municipality profile interface, deploy national and provincial visualization dashboards, activate high-risk radar components, open public search channels, and launch the basic public API tier.
- **Outcome:** A live, demonstration-ready accountability application and operational platform prototype.

14.3 Phase 2: Functional Depth Expansion

- **Deliverables:** Integrate real-time financial tracking modules, deploy infrastructure project tracking features, activate metropolitan comparative tools, launch automated briefing export functions, and roll out enterprise api subscription access.
- **Outcome:** Formal commercial launch, backed by integrated live financial metrics and automated report delivery options.

14.4 Phase 3: Operational Scale

- **Deliverables:** Implement custom user watchlists and notification triggers, activate high-availability enterprise service tiers, ingest deep historical records, and introduce advanced workflow management options.
- **Outcome:** A mature, self-sustaining civic tracking platform and analytical standard.

15. Appendix A: Comprehensive Glossary

15.1 Audit Classification Terms (AGSA)

- **Clean Audit:** An audit outcome confirming a municipality's financial statements are fairly presented and comply with regulatory guidelines, with no material findings regarding institutional performance or legal processes.
- **Unqualified Opinion with Findings:** An outcome indicating that while a municipality's financial statements are fairly presented, auditors identified material compliance exceptions, process failures, or performance monitoring gaps.
- **Qualified Opinion:** An audit finding stating that while the overall financial records are mostly sound, specific account structures contained material misstatements or lacked sufficient documentation for verification.
- **Adverse Opinion:** A critical audit outcome indicating that a municipality's financial statements contain widespread misstatements that do not accurately reflect its financial position.
- **Disclaimed Opinion:** The lowest audit classification, issued when a municipality fails to provide sufficient records for auditors to form an opinion on its financial statements.
- **Material Irregularity (MI):** Any systemic compliance failure, fraud attempt, or administrative neglect identified during an audit that results in material financial losses or compromises essential public service delivery.
- **Irregular Expenditure:** Municipal spending that diverges from statutory procurement rules and public management guidelines, but does not automatically imply fraud.
- **Unauthorised Expenditure:** Administrative spending that exceeds approved budget votes or falls outside the council's legislative mandates.
- **Fruitless and Wasteful Expenditure:** Direct public expenditure incurred in vain that could have been avoided by applying reasonable care and standard internal checks.
- **Certificate of Debt:** A formal enforcement instrument issued by the Auditor-General that holds an official personally liable for returning lost public funds resulting from clear systemic neglect.

15.2 Financial Telemetry Cube Identifiers (National Treasury API)

- `incexp` (**Income & Expenditure Cube**): The operational data stream tracking incoming revenue collections alongside active monthly operational spending.
- `cflow` (**Cash Flow Cube**): The data stream tracking absolute cash movements, monitoring short-term liquidity, and tracking operational cash variations.
- `bsheet` (**Balance Sheet Cube**): The structural database ledger detailing municipal assets, liabilities, and long-term fund balances.
- `capital` (**Capital Acquisition Cube**): The data path tracking long-term structural assets, evaluating capital project allocations, and monitoring infrastructure renewals.
- `conditional_grants` (**Conditional Grants Cube**): The reporting stream tracking specialized national grant allocations (e.g., *MIG*, *USDG*), monitoring quarterly drawdowns, and identifying unspent funds at risk of reallocation.